PRASANTHKUMAR PONNADA

Machine Learning | Data Science | Python

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience in Python, Machine Learning, NLP, and Generative AI, with a strong academic foundation in Artificial Intelligence. Experienced in developing end-to-end ML solutions including customer sentiment analysis, CSAT prediction, topic modeling, recommendation systems, and AI-powered applications. Proficient in developing, evaluating, and deploying machine learning models with a focus on practical business applications.

TECHNICAL SKILLS

- **Programming**: Python, NumPy, Pandas, Scikit-learn, SQL
- **Machine Learning**: XGBoost, Random Forest, Classification, Model Evaluation, Feature Engineering
- **NLP & Gen AI**: BERT, Transformers, VADER, LDA, Coqui TTS (XTTS/VITS)
- **Data Science**: Data cleaning, Statistical Analysis
- **Deployment & Tools**: Git, GitHub, Gradio, Hugging Face Spaces, AWS S3/EC2, Matplotlib, Plotly

PROJECTS

Customer Sentiment & CSAT Engine for Airlines

GitHub: https://github.com/prasanthppk/Customer_Sentiment_Analysis.git

Python | NLP | NLTK | Scikit-Learn | BERT | LDA | Gradio | Hugging Face

- Built end to end pipeline processing 1500+ passenger feedback records (text and structured ratings)
- Engineered sentiment features using VADER/BERT, trained classifier for CSAT (High/Low/Neutral) with ~90% accuracy.
- Built LDA topic modelling recommendation engine to map issues (Refund, Food, Delay, etc.,) to business actions.
- Deployed interactive Gradio app on Hugging Face Spaces.

Telecom Customer Churn Prediction

GitHub: https://github.com/prasanthppk/Customer_Churn_Prediction.git

Python | Scikit-Learn | Pandas | NumPy | Gradio | Hugging Face

- Developed Machine Learning Model to predict customer churn using customer demographics, service usage and billing information.
- Performed Data preprocessing, EDA, Feature engineering and evaluated 3 algorithms. Best: XGBoost with 92.5% F1-Score.
- Identified key churn driving factors and generated predictions to help telecom companies implement targeted customer retention strategies.

Voice Cloning Application using Coqui TTS

Python | Coqui TTS configured models | VITS | XTTS

- Developed a voice cloning application using Coqui TTS to generate natural sounding speech that replicates a target speaker's voice characteristics.

- Implemented audio preprocessing and voice sample handling to prepare speaker recordings for cloning and speech synthesis.
- Integrated text-to-speech pipeline that converts user-provided text into synthesized speech using the cloned voice.

Deep Learning Based Fashion Item Classification using Fashion-MNIST

Python | NumPy | TensorFlow | Scikit-Learn | Keras | CNN | Matplotlib

- Developed an image classification model to automatically identify fashion items form the Fashion MNIST dataset, covering categories such as shirts, shoes, bags and trousers.
- Performed image preprocessing and normalization to prepare grayscale image data for efficient model training.
- Implemented and trained a neural network-based classifier to learn visual patterns and classify fashion images into their respective categories.
- Evaluated model performance using accuracy, loss, confusion matrix and classification metrics to assess classification effectiveness.

PROFESSIONAL EXPERIENCE

Senior Analyst | Satyam Venture Engineering Services, Hyderabad | Feb 2022 - Present

- Developed and reviewed simulation models while ensuring accuracy and adherence to project requirements.
- Building Python automation to process large scale simulation datasets to reducing manual analysis time.
- Developing data-driven models for failure prediction and collaborated with cross functional teams.
- Awarded Star of the Month recognition for error-free deliveries.

CAE Engineer | Xitadel CAE Technologies India Pvt. Ltd., Bangalore | Nov 2018 – Jan 2022

EDUCATION

- **Associate of Science** in AI | Saras AI Institute, Salt Lake City, UT, USA. | CGPA: **3.5 / 4** | Relevant: ML, DL, NLP, GenAI, DSA, Statistics, Probability
- **M.Tech** Thermal Engineering | Gayatri Vidya Parishad College of Engineering, Visakhapatnam, A.P. | **69.17%**
- **B.Tech** Mechanical Engineering | Gudlavalleru Engineering College, Krishna, A.P. | **72.15%**

CERTIFICATIONS

Machine Learning – [Saras AI Institute] | Python for Data Science – [Saras AI Institute] | AWS Cloud Practitioner

Date: 30 October, 2026

Place: Hyderabad [Prasanthkumar Ponnada]